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Project Title:	Habitability and Human Factors Assessment in CHAPEA (iSHORT, SHAQ and SHU)		
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	As long-duration off-planet habitats become a reality, a consideration of habitability and human factors (HF) is crucial. The habitat is more than just a place to live and work. It is also the crew's perception of the space, and the psychological impacts of size, layout, and usage over time; all of which can support or strain behavioral health and performance (BHP). The Crew Heath and Performance Exploration Analog (CHAPEA) study is a Mars habitat simulation in which we examine the human health and performance of crews living and working in isolation for a year with Mars realistic resource restrictions including a limited food system and communication delay with mission control. iSHORT (Space Habitability Observation Reporting Tool) was previously used to collect detailed data about habitability and human factors on the International Space Station (ISS) to inform NASA Standards (see Greene, Thaxton, & Adolf, 2018, report to NASA Human Factors & Behavioral Performance Element) and to demonstrate the iSHORT tool. Only one of the six ISS iSHORT study subjects had a duration of one year; all other ISS and ground analog subjects had mission durations		

Task Description:	from one week up to six months. CHAPEA data will be collected during three, one-year missions to gather needed long-duration data related to habitability. iSHORT will also include semi-structured prompts to elicit crewmembers' reflections on key concerns of human factors design and behavioral health. The data collected with the iSHORT will be compared to and complement data collected with the SHAQ (Subjective Habitability and Acceptability Questionnaire; Roma, Landon, et al., 2022), which was deployed in CHAPEA to examine the intersection of psychology and habitat design. Additionally, this mission allows human factors experts to field test a new tool, the Scale for Habitat Usability (SHU), which is the first step for transitioning to operations. The SHU is a brief subjective scale which captures important elements of how habitat design impacts task performance. The aim of the Habitability and Human Factors Assessment (HHFA) study in CHAPEA is to capture HF design concerns and related BHP impacts in a high-fidelity spaceflight analog. iSHORT captures crewmembers' thoughts, positive and negative, about different habitat points of interest (POI) (i.e., habitat areas, activities, key equipment). Crew will reflect on each habitat POIs multiple times during the mission, which enables better understanding of the change in acceptability over time. We will compare results between the 3 mission crews to understand how individual well-being and team dynamics may be related to human factors concerns over time. New data collected in CHAPEA will also inform human factors design and NASA Standards. Mission 1 began in June 2023 and completed in July 2024. Mission 2 is scheduled for 2025-2026.
Rationale for HRP Directed Research:	This proposal qualifies for a directed study due to the urgent, time-sensitive need to provide "standard measures" as the foundation to achieve consistent research measures and to meet the highly constrained, operationally focused data gathering and analysis that allows for greater consistency in the research methods that are very specific to NASA Human Research Program (HRP) standard measures development. The comparison and complementary understanding of the 3 habitability and human factors measures in this study will allow efficient implementation of the measures in analogs and/or spaceflight in near-term research. Additionally, the directed nature of this study also allows the BHP Laboratory and the Human Factors Engineering Laboratory to provide the unique research and support expertise required to integrate and manage the data from the various participating studies to achieve HRP's intent. Access to the BHP Laboratory's HFBP-Exploration Measures (EM) database and vetting of the evidence-based standards makes the solicitation process prohibitive.
Research Impact/Earth Benefits:	Information related to habitability in isolated, confined, extreme, and operational spaces over time can be used by Earth-oriented habitat designers to improve well-being, social relationships, and performance.
Task Progress:	Data Collection. Both individual and team-level data collections were repeated at regular intervals throughout all three mission phases (pre-, in-, and post-). Data was collected using three measures (iSHORT, SHAQ, and SHU), along with habitability-focused team meetings, and a post-mission team debriefs. Individual level. Crew completed surveys and provided video and audio recording observations and entries on several habitat areas, activities, and key equipment (i.e., points of interest/POIs). These points of interest include the kitchen/galley, crew quarters, and other work and living areas. iSHORT is an app with an accompanying survey which allows for novel, subject-curated observations on each POI through its think-aloud protocol which asks users to provide video and audio recordings and different areas and include tags on considerations and activities depending on POI use and description. The iSHORT standalone survey also includes the SHU which focuses on usability. The SHAQ survey includes evaluations of privacy, comfort, convenience, control, efficiency, and social density through the lens of subsequent outcomes like sleep, individual performance, group activities performance, stress, mood, and social interactions. Pre- and post-mission evaluations will also allow comparison with homes, pre- and post-mission hotels, and a retrospective reflection of living and working in a long-duration analog. Team level. Throughout the mission, the crew had a weekly team meeting to discuss various topics. The crew had six habitability-focused crew meetings, once every 9 weeks, while in-mission. These round table discussions are semi-structured feedback sessions in which crewmembers work through habitability-oriented design tasks together. Two meetings focused on the CHAPEA habitat layout and were conducted in Mission Weeks 5 and 53 to examine changes over time. Four focused on furniture, team rationale, and decision-making. These meetings were further subset into Living Areas and Working Areas. Both topics were discussed twice by crew to compare the first half of the mission to the second half of the mission. Teams were asked how they might improve these aspects of the habitat and furniture for their needs, and they also recorded a brief statement about their rationale and team decision-making process. The tasks inform our habitability objectives by offering insights into team rationale and the team decision-making processes with regard to the habitat over time. Data Yield. The research team achieved 100% data yield across all three mission phases. Analysis of survey data and transcriptions of individual and team activities is underway. Pre-Mission. During the Crew Simulation Day, in which the crew completed several tasks in the habitat to test the facility and various operations and research systems, the iSHORT was offered as part of the crew's simulated mission activities. The research team implemented the iSHORT post-mission protocol with a Profile of Mood Survey Short Form (POMS-SF) and SHU. In-Mission. The iSHORT Standalone was collected three times for three different POIs, SHAQ Standard had six collections for ten POIs, and iSHORT with SHAQ had three collections for ten POIs. Data was also collected during six in-mission team meetings focused on habitability and human factors. Post-mission. The crew completed a post-mission iSHORT with seven POIs and the SHAQ survey about their hotel. A debrief with the entire CHAPEA Mission 1 crew allowed the crew to provide feedback on the acceptability and cadence of the in-mission habitability measures.
Bibliography Type:	Description: (Last Updated: 01/12/2026)
Abstracts for Journals and Proceedings	Miller JCW, FarrHenderson M, Robertson IWT, Bell ST, Landon LB. "Habitability and human factors assessment (iSHORT, SHAQ, and SHU)." 2024 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 13-16, 2024. Abstracts. 2024 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 13-16, 2024. , Feb-2024